

Centralized Microgrid Controller with FS-MPC Interface Converters and Digital Twin Integration: Implementation Plan

Anoop Eluvathingal

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Centralized Microgrid Controller with FS-MPC Interface Converters and Digital Twin Integration — Implementation Plan

Author: Anoop Eluvathingal (IIT Madras / NUS / NTU) **Date:** 26 April 2026 **Purpose:** Engineering plan to scale the Perez & Flores-Bahamonde 2016 single-converter FS-MPC reactive/harmonic compensation scheme into a fleet of microgrid interface converters governed by a centralized microgrid controller (CMC), validated against and continuously calibrated by a microgrid digital twin (MG-DT). Direct scaffolding for MAS-DT-SH thesis Sprints 3–6 and the APPEEC 2026 paper #2.

0. Executive Summary

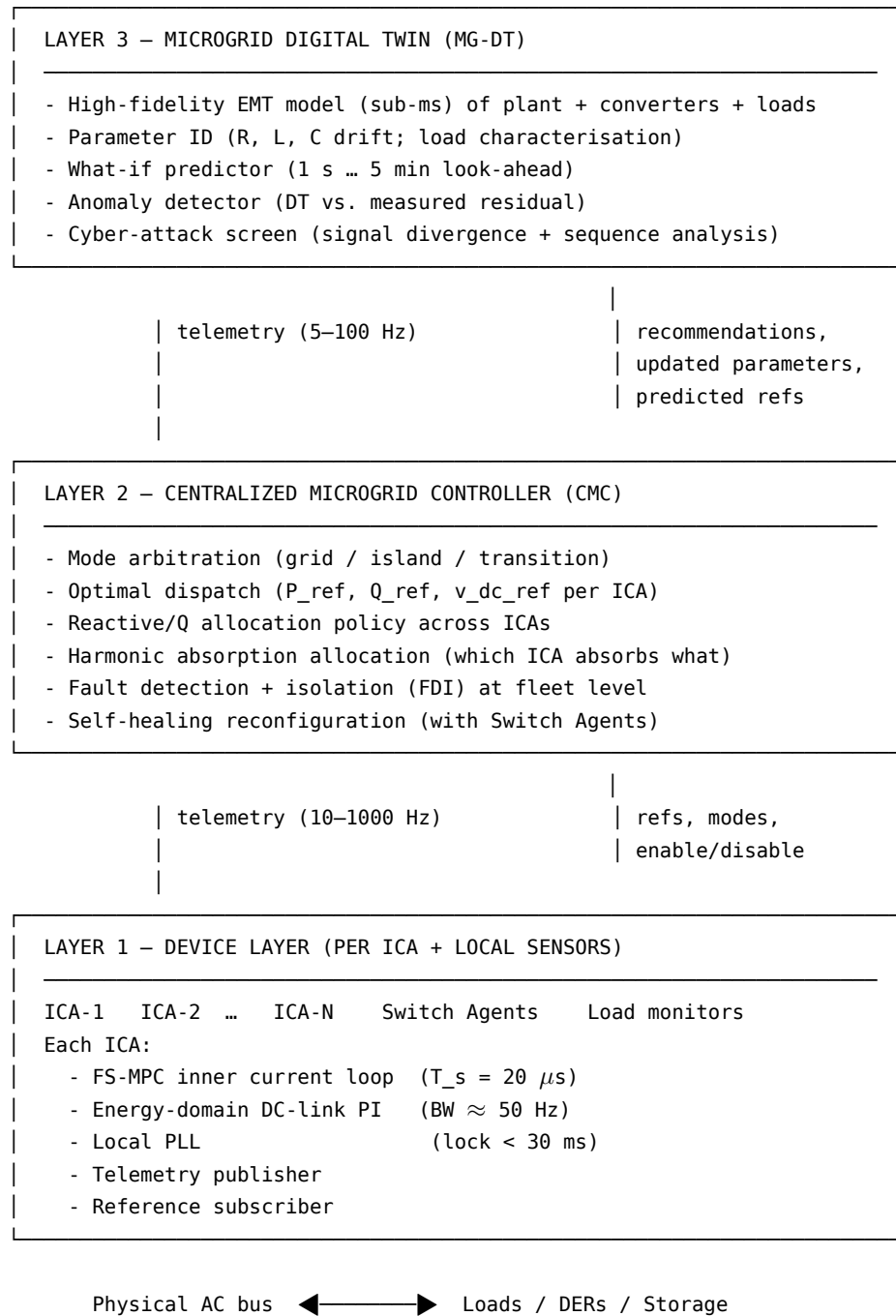
The single-converter FS-MPC scheme of [F] gives a microgrid interface converter (MIC) the ability to act simultaneously as a bulk-power gateway and a power-quality device. Operating *one* such converter is well-understood. Operating *several* of them as a coordinated fleet, while a digital twin runs in parallel checking, calibrating, and forecasting, is the open problem this plan addresses.

The proposed architecture is three-layered: a **device layer** of per-converter FS-MPC + outer energy-PI controllers (Layer 1, μs –ms timescale); a **centralized microgrid controller (CMC)** layer that arbitrates mode, dispatches references, and detects faults (Layer 2, s–min timescale); and a **microgrid digital twin (MG-DT)** layer that mirrors the physical plant in soft real-time and provides four services to Layers 1 and 2 — online parameter ID, predictive what-if, anomaly detection, and cyber-attack screening (Layer 3, soft real-time, ms–s).

The plan is staged across **four phases over twelve months** (Single ICA $\rightarrow$ Fleet $\rightarrow$ DT integration $\rightarrow$ HIL validation), with twenty-six concrete deliverables, a 12-test integration matrix, and risk register. Total estimated effort: **140 working days** for one PhD candidate, parallelisable to ~ 85 calendar days with one MS-thesis collaborator on the DT side.

Part A — System Architecture

1. Three-Layer View



2. Layer Responsibilities — Functional Allocation

Function	Layer 1 (Device)	Layer 2 (CMC)	Layer 3 (DT)
Switching-state selection	✓ FS-MPC at 50 kHz	—	—
Inner current ref tracking	✓	—	shadow-runs
DC-link voltage regulation	✓ outer PI	sets v_dc_ref	shadow-runs
Reactive power dispatch	local Q at PCC	✓ optimal Q split	predicts load Q
Harmonic absorption split	local capability	✓ allocation policy	identifies harmonic profile

Function	Layer 1 (Device)	Layer 2 (CMC)	Layer 3 (DT)
Mode (grid/island) handling	✓ executes	✓ decides	predicts transition
FDI / self-healing	local sensor only	✓ system-wide FDI	early-warning
Plant parameter calibration	uses values	applies updates	✓ identifies online
Cyber-attack detection	none	basic plausibility	✓ residual analysis
Operator dashboard	—	live KPIs	✓ what-if scenarios
Economic dispatch (tertiary)	—	✓ slower MPC	input forecast

3. Latency Budget

Path	Period / latency target	Why
Sensor → FS-MPC predictor	$< 5 \mu\text{s}$	within $\tau_s = 20 \mu\text{s}$ budget
FS-MPC inner loop control	$20 \mu\text{s}$	fixed by paper [F]
Outer energy PI	$200 \mu\text{s}$ (5 kHz)	aliased control
ICA → CMC telemetry publish	10 ms (100 Hz)	suffices for sub-cycle aggregate metrics
CMC → ICA reference push	10 ms	bound by Ethernet RTT + serialisation
CMC dispatch optimisation	1 s	tertiary-style economic + Q-allocation MPC
DT real-time tick	100 μs internal, 10 ms publish	EMT fidelity for residuals
DT parameter-ID update	10 s	low-rate calibration
Mode-transition decision	$< 100 \text{ ms}$	IEEE 1547-2018 RT requirements
Fault isolation (CMC + Switch Agents)	$< 200 \text{ ms}$	industry expectation
Cyber-attack alarm	$< 1 \text{ s}$	usable SOC alert

These numbers feed the protocol selection in Part E.

Part B — Layer 1: Per-Converter Device Software (Interface Converter Agent)

4. ICA Module Map

ICA Agent

└─ core/	
└─ plant.py	# Eq.(1)(2) state model – used by predictor + DT
└─ inner_fsmpc.py	# 8-vector predictor + cost + argmin + delay comp
└─ outer_energy_pi.py	# Energy-domain PI on $E_c = 1/2 C v_{dc}^2$
└─ pll.py	# SOGI-PLL positive-sequence detector
└─ modes.py	# local FSM: starting / running / fault / mode_switch
└─ io/	
└─ adc.py	# measurement abstraction (HW or sim)
└─ pwm.py	# gate driver abstraction
└─ safety.py	# over-current, over-voltage hard cuts (independent of FS-MPC)
└─ comm/	
└─ telemetry.py	# publishes at 100 Hz to CMC and DT
└─ refs.py	# subscribes to v_{dc_ref} , P_{ref} , Q_{ref} , mode
└─ time_sync.py	# PTP/IEEE 1588 interface
└─ selftest/	
└─ unit_predictor.py	# property tests on Eq.(4)
└─ thd_meter.py	# online FFT-based THD (post-control)
└─ audit_log.py	# ring buffer of switching states for forensics
└─ main.py	# event loop (RT priority)

5. Inner-Loop Pseudocode

```

loop every T_s = 20 μs:
    measure i_m, v_dc, v_s, i_l
    v_s_pos, theta = pll.update(v_s)

    # outer loop runs slower; latest setpoint is cached
    I_s_amp = outer.last_amplitude
    i_s_ref = I_s_amp * sin(theta)
    i_m_ref = i_s_ref - i_l

    best_s, best_cost = None, +inf
    for s in 8_switching_vectors:
        i_pred = (1 - r*T_s/L) * i_m + (T_s/L) * (v_s - M @ s * v_dc)
        cost = sum((i_pred - i_m_ref)**2) # 3-phase squared error
        if cost < best_cost:
            best_s, best_cost = s, cost

    pwm.apply(best_s) # one-step delay compensated separately
    telemetry.append((t, i_m, v_dc, best_s, cost))

```

6. Outer-Loop Pseudocode

```

loop every 200 μs:
    E_c = 0.5 * C * v_dc**2
    E_c_ref = 0.5 * C * v_dc_ref**2
    err = E_c_ref - E_c
    integ += err * dt
    I_s_amp = Kp*err + Ki*integ + I_s_ff # feedforward of (p_dc + p_l)
    outer.last_amplitude = I_s_amp

```

7. Reference Interface (CMC → ICA)

Topic	Type	Rate	Description
/ica/{id}/refs/v_dc_ref	float	1 Hz	DC-link target
/ica/{id}/refs/Q_ref	float	10 Hz	reactive support
/ica/{id}/refs/H_mask	float[H]	1 Hz	harmonic absorption priority per order
/ica/{id}/refs/mode	enum	event	grid/island/idle
/ica/{id}/refs/enable	bool	event	safety arm

Reactive Q_{ref} is converted locally into an I_{s_ref} amplitude phase shift; harmonic mask H_{mask} filters which orders the FS-MPC compensates.

Part C — Layer 2: Centralized Microgrid Controller

8. CMC Functional Decomposition

```

CMC
├─ state_estimator/
│   ├── topology_tracker.py # graph of buses, switches, ICAs, loads
│   ├── pseudomeasurement.py # fills missing telemetry from DT
│   └─ fault_detector.py # residual-based FDI
├─ optimizer/
│   ├── mode_arbiter.py # FSM grid/island/transition
│   ├── q_allocator.py # OPF-style Q allocation across ICAs
│   ├── h_allocator.py # harmonic-absorption splitter per order
│   └─ dispatch_mpc.py # tertiary-style economic MPC (1 min horizon)

```

```

├─ self_healing/
│   ├── isolator.py          # commands to Switch Agents
│   ├── reconfiguration.py   # post-fault topology
│   └─ black_start.py        # islanded re-energisation sequence
├─ operator/
│   ├── dashboard_ws.py      # live KPI websocket
│   ├── manual_overrides.py  # operator inputs
│   └─ alarms.py             # event log
├─ api/
│   ├── ica_pubsub.py        # tx refs, rx telemetry
│   ├── dt_bridge.py         # bidirectional with DT
│   └─ scada_dnp3.py         # interface to wider grid SCADA
└─ main.py

```

9. Q Allocation Optimisation (Heart of CMC)

At each control tick (1 s), CMC solves a small QP:

```

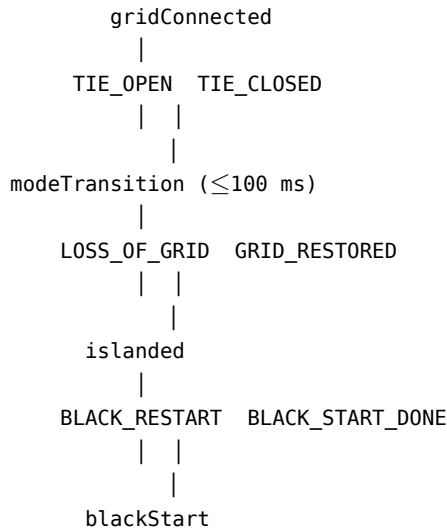
minimize    sum_i α_i * Q_i^2 + β * sum_i (V_i - V_ref)^2
            + γ * (sum_i Q_i - Q_total_target)^2
subject to  |Q_i| ≤ S_i^max   (apparent-power limit per ICA)
            Q_i in feasible region given P_i (capability curve)
            sum_i P_i = P_load + P_loss   (active power balance)
            grid-tie limits if grid-connected

```

Q_{total_target} is set by the secondary voltage controller; α_i, β, γ are weights tuned offline. The output Q_i is broken into a *fundamental* part (sent as Q_{ref}) and a *harmonic-absorption capacity* (sent as H_{mask}).

Why a slow MPC and not droop? In a centralized architecture with reliable comms, the QP gives globally optimal Q sharing rather than the local-only droop optimum. When comms degrade, the CMC falls back to droop coefficients pushed to ICAs — graceful degradation.

10. Mode Arbitration FSM



CMC owns the FSM. ICAs are told the mode but execute autonomously (mode-agnostic FS-MPC of [F] simplifies the transition).

11. Harmonic Allocation (Novel Contribution)

In a fleet of N ICAs, total harmonic load i_{l_h} ($h = 5, 7, 11, 13\dots$) can be split among them in proportion to their *available headroom* (peak current limit minus fundamental current):

$$H_{i_h} = i_{l_h_total} * (S_i^{max} - |I_{i_fund}|) / \sum_j (S_j^{max} - |I_{j_fund}|)$$

The `h_allocator.py` publishes H_{mask} to each ICA. This is the contribution that turns Perez’s single-converter scheme into a coordinated fleet — an obvious extension to claim in your thesis chapter.

Part D — Layer 3: Microgrid Digital Twin

12. DT Architecture

MG-DT

```

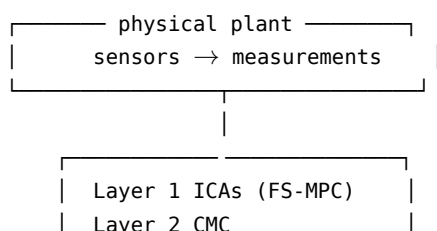
├─ physics/
│   ├── emt_model.py          # OpenModelica / PSCAD / Python EMT – sub-ms step
│   ├── grid_eq.py           # equivalent grid (Thevenin or PQ profile)
│   ├── load_models.py       # ZIP, CIGRE-aggregated, 6-pulse rectifier presets
│   └── icas_replica.py      # mirrors of all ICAs
├─ identification/
│   ├── recursive_ls.py      # online R, L, C update from residuals
│   ├── load_classifier.py   # classifies load harmonic profile
│   └── parameter_kalman.py  # Kalman state-parameter joint estimator
├─ prediction/
│   ├── horizon_simulator.py # rolls forward 1–300 s with stochastic loads
│   ├── what_if.py          # branched scenarios
│   └── fault_propagator.py  # injects faults to predict CMC response
├─ analytics/
│   ├── residual_monitor.py  # measured – DT prediction
│   ├── anomaly_detector.py  # one-class SVM on residuals
│   ├── cyber_screen.py     # message-level integrity + signal divergence
│   └── ml_advisor.py       # offline-trained NN for FS-MPC weight hints
├─ api/
│   ├── cmc_bridge.py        # CMC ↔ DT (send predictions, receive snapshots)
│   ├── ica_shadow.py       # parallel ICA emulators
│   └── operator_view.py    # 3D viz endpoint
└─ orchestrator.py

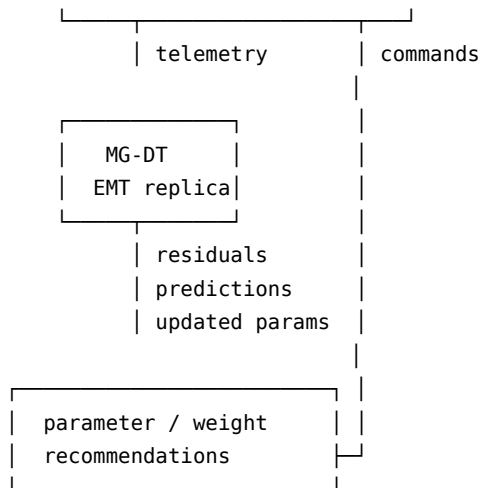
```

13. DT Use Cases (5 prioritised)

#	Use case	Direct benefit	Layer
U1	Online plant-parameter ID for FS-MPC predictor	improves <code>i_m_pred</code> accuracy under R, L drift	feeds Layer 1
U2	Look-ahead Q forecast from load classifier	CMC pre-emptively dispatches Q	feeds Layer 2
U3	Anomaly residual between DT and measured	early sensor-fault detection	Layer 2 alert
U4	Cyber-attack screening on signal/message integrity	rejects spoofed <code>Q_ref</code> injections	Layer 2 alert
U5	Operator what-if for proposed actions	“what if we open Tie-2 at 10:30?” simulated	Layer 2 UI

14. DT–Controller Co-design Loop





The closed feedback from DT into both Layer 1 (parameters) and Layer 2 (predictions, Q forecasts, what-ifs) is the cyber-physical loop that distinguishes a true digital twin from a plain simulator.

15. DT Realisation Choices

Aspect	Recommendation	Alternatives
EMT engine	OpenModelica + Python orchestration via FMI	PSCAD (license), Simulink, Typhoon HIL
State exchange	FMI-Cosim 2.0	OPC-UA, gRPC streams
Parameter ID	RLS + Kalman hybrid	particle filter, neural ID
Anomaly detector	One-class SVM + hard threshold	LSTM autoencoder
Cyber screen	Hash-chained sequence numbers + STL on signals	dedicated SIEM
Real-time host	Linux PREEMPT-RT + Docker	bare-metal RTOS

Part E — Software Stack & Deployment

16. Languages and Frameworks

Layer	Primary	Reason	Secondary
Layer 1 (ICA)	Python 3.11 + numpy + numba for the simulation/prototype; C++/HLS when porting to FPGA/DSP	matches <code>fs_mpc_microgrid</code> repo from the forensic plan	MATLAB/Simulink coder for industry validation
Layer 2 (CMC)	Python 3.11 + cvxpy + asyncio	QP solver + async I/O	Julia/JuMP if larger systems
Layer 3 (DT)	OpenModelica + Python orchestrator	open EMT, FMI-friendly	Simulink + Simscape Electrical
Operator UI	React + d3 over websocket	live dashboards	Grafana

17. Communication Protocols

Path	Protocol	Why
Sensor → ICA	analog/digital local	local DSP I/O
ICA ↔ CMC	MQTT v5 (publish/subscribe) over TLS	low-latency, broker-friendly
ICA ↔ DT shadow	DDS (RTI Connex or eProsima Fast DDS)	RT pub/sub with QoS
CMC ↔ DT	OPC-UA + FMI	industrial standard, vendor-neutral
Switch Agents	IEC 61850 GOOSE	sub-4 ms tripping
External SCADA	DNP3 / IEC 60870-5-104	utility integration
Operator UI	WebSocket + JSON	browser-friendly

18. Containerisation and Deployment

docker-compose.yml

```

|— ica-{1..N}      # one container per ICA agent (or per real device)
|— cmc             # central controller
|— mg-dt           # digital twin
|— mqtt-broker     # mosquitto
|— dnp3-gateway    # external interface
|— ui-frontend     # React app
|— postgres        # event log + telemetry archive

```

In production each container is pinned to a CPU core (PREEMPT-RT) and the whole stack runs on a 1 RU industrial PC (Moxa V2616A or equivalent). For HIL the CMC runs on the host while the ICAs run inside a Typhoon HIL or OPAL-RT.

Part F — Test Plan

19. Test Pyramid

Level 4: Field pilot (after Phase 4)

Level 3: Hardware-in-loop (Typhoon HIL + 2 real ICAs)

Level 2: System integration (full Docker stack, simulated plant)

Level 1: Component integration (ICA + CMC, ICA + DT)

Level 0: Unit tests (per module, per loop)

20. Test Matrix (12 cases)

ID	Level	Scenario	Pass criterion
T01	0	FS-MPC predictor unit (Eq. 4)	tracks step i_{m_ref} in 1–2 samples
T02	0	Outer PI energy loop	v_{dc} settles in 20 ± 5 ms on 100 A step
T03	0	SOGI-PLL on distorted v_s	locks within 30 ms, $\pm 2^\circ$ error at 5% THD
T04	1	ICA standalone — reproduce [F] Fig 4–8	THD $i_s < 5\%$ in all 3 modes
T05	1	CMC Q-allocator vs. droop baseline	$\geq 10\%$ lower total $ Q $ for same V error
T06	1	DT parameter-ID convergence on R drift	2% accuracy in < 10 s

ID	Level	Scenario	Pass criterion
T07	2	4-ICA fleet — coordinated harmonic absorption	system <code>i_s</code> THD < 3% per ICA, < 2% PCC
T08	2	Mode transition grid → island in 100 ms	$\backslash v_{dc}\backslash$ deviation < 5%, no trip
T09	2	DT-flagged anomaly during sensor fault on ICA-2	DT alarm < 500 ms, CMC isolates ICA-2
T10	2	Cyber screen rejects spoofed <code>Q_ref</code>	rejected within 1 s, audit log entry
T11	3	HIL: 2 real + 2 simulated ICAs	THD < 5%, settling within spec, no oscillation
T12	4	Pilot site (small commercial)	7-day continuous run, reactive billing reduction $\geq 30\%$

Part G — Implementation Roadmap

21. Phase Plan (12 months)

Month	1	2	3	4	5	6	7	8	9	10	11	12
Phase 1 :	■	■	■	■	■	■	■	■	■	■	■	■
Phase 2 :			■	■	■	■	■	■	■	■	■	■
Phase 3 :				■	■	■	■	■	■	■	■	■
Phase 4 :						■	■	■	■	■	■	■

Single ICA reproduce + harden (T01–T04)

Fleet + CMC (T05, T07, T08)

DT integration (T06, T09, T10)

HIL + pilot (T11, T12)

22. Phase 1 — Single ICA (Months 1–3, 35 person-days)

#	Deliverable	Day
D01	Repo scaffold <code>fs_mpc_microgrid/</code> (per earlier forensic plan)	1
D02	Plant model <code>plant.py</code> + tests	3
D03	6-pulse + linear load model	5
D04	SOGI-PLL implementation + lock test	7
D05	FS-MPC inner loop + delay-comp + tests	12
D06	Outer energy PI + tests	14
D07	Reproduce [F] Fig 4–10 in notebooks	20
D08	THD quantification + report	22
D09	ICA agent wrapper (<code>io.adc</code> , <code>comm.refs</code> , <code>comm.telemetry</code>)	28
D10	First end-to-end ICA-only Docker container	35

23. Phase 2 — Centralized Controller + Fleet (Months 4–6, 40 person-days)

#	Deliverable	Day
D11	MQTT broker + topic schema	38
D12	CMC scaffold <code>cmc/</code> repo	40
D13	Topology tracker + state estimator	47
D14	Mode arbitration FSM + tests	52
D15	Q allocator (<code>cvxpy</code> QP) + tests	58
D16	Harmonic allocator + tests	62
D17	4-ICA fleet simulation harness	70
D18	Operator dashboard MVP	75

24. Phase 3 — Digital Twin (Months 7–9, 35 person-days)

#	Deliverable	Day
D19	OpenModelica EMT replica	80
D20	FMI bridge to CMC	85
D21	RLS parameter ID	92
D22	Anomaly detector + cyber screen	100
D23	Look-ahead Q forecaster	105
D24	DT-CMC closed loop + integration tests T06, T09, T10	110

25. Phase 4 — HIL + Pilot (Months 10–12, 30 person-days)

#	Deliverable	Day
D25	Typhoon HIL or OPAL-RT integration	120
D26	Two-real-ICA hardware bench (NUS PSCL or IIT Madras lab)	130
D27	T11 HIL test pass	135
D28	T12 pilot deployment	140

26. Effort Summary

Phase	Person-days	Calendar months
Phase 1	35	3
Phase 2	40	3
Phase 3	35	3
Phase 4	30	3
Total	140	12

With a single MS-thesis collaborator on the DT side (Phase 3 in parallel with Phase 2 final third), calendar can compress to ~9 months.

Part H — Risks and Mitigations

Risk	Probability	Impact	Mitigation
FS-MPC Forward Euler instability at $T_s = 20 \mu s$ and $L = 1 mH$	Medium	High	Use trapezoidal predictor; verify pole locations
Hardware MQTT broker latency exceeds 10 ms budget	Medium	Med	Shadow with DDS or co-locate broker on CMC host
Anomaly false alarms swamp operator	High	Med	Tune SVM threshold offline with historical residuals; gating with multi-sample window
OpenModelica + Python FMI sync drift	Medium	Med	Use FMI-Cosim 2.0 with Jacobi master, audit reconciliation step
Data-driven param ID overfits transient	Medium	Med	Frequency-domain bandlimit on RLS update; freeze during transitions

Risk	Probability	Impact	Mitigation
Cyber screen rejects legitimate operator override	Low	Med	Operator authentication path (signed tokens) bypasses signal screen
Loss of comms degrades CMC dispatch	High	High	Local droop fallback already loaded in each ICA
HIL availability bottleneck	High	Med	Reserve Typhoon early (Month 8), have Python EMT fallback
Phase-3 (DT) requires more domain knowledge than budgeted	Medium	High	Recruit MS-thesis collaborator with Modelica experience
Pilot site permission delays	Medium	Med	Early engagement with NUS / IIT Madras facilities team in Month 6

Part I — Master Summary Table

Aspect	Choice / Value
Architecture	3-layer: Device · CMC · MG-DT
Inner loop	FS-MPC, $T_s = 20 \mu s$, 8 vectors, delay-comp at $k+2$
Outer loop	Energy-domain PI on $E_c = (1/2)Cv_{dc}^2$
Centralized dispatch	1 Hz QP for Q + harmonic allocation
DT engine	OpenModelica EMT + Python FMI orchestrator
DT services	Parameter ID · Q forecast · Anomaly · Cyber screen · What-if
Comms	MQTT v5 (ICA↔CMC), DDS (ICA↔DT), OPC-UA + FMI (CMC↔DT), GOOSE (Switches)
Languages	Python (Layers 1, 2, DT orchestrator), Modelica (DT physics)
Container	Docker + PREEMPT-RT Linux on industrial PC
Test pyramid	12-case matrix from unit to pilot
Phases	4 phases × 3 months each
Total effort	140 person-days; ≥ 9 calendar months with collaborator
PhD relevance	Maps directly onto MAS-DT-SH Sprints 3 (orchestrator), 4 (switch agents), 5 (taxonomy), 6 (Algorithm-1 end-to-end)
Patent angle	Harmonic-allocation policy across ICAs (Section 11)
Publication anchor	Reproduction + extension paper to IEEE Trans. Smart Grid 2026

Part J — Concrete Next-Week Tasks

If you start tomorrow (Mon 27 Apr 2026):

Day	Task	Output
Mon	Create <code>fs_mpc_microgrid/</code> repo, <code>pyproject.toml</code> , CI	green pipeline
Tue	Implement <code>plant.py</code> Eq. (1)(2)	unit tests pass
Wed	Implement <code>inner_fsmpc.py</code> Eq. (4)	step response visible
Thu	Implement <code>outer_energy_pi.py</code>	DC-link converges to ref
Fri	First ICA simulation reproduces [F] Fig 4	THD < 5%
Sat	Capture screenshots, draft thesis-Ch4 paragraph + APPEEC #2 citation	text drafted
Sun	Sabbath — review the week, lock Phase 1 plan	decision log entry

This single week converts the present plan from intent into evidence and unblocks Sprint 2 of MAS-DT-SH while you're in the APPEEC submission home stretch.

Companion documents on the same workspace folder: - `FS_MPC_Microgrid_Forensic_Analysis.md` — line-by-line read of the focal paper [F]. - `FS_MPC_Related_References_From_Drive.md` — Tier-1 reference library curation. - `FS_MPC_Microgrid_Literature_Review_IEEE.pdf` — full systematic review with figures. - `~/Desktop/fs_mpc_refs/` — local Tier-1 PDFs for offline reading.